小目标评价指标

指标	评价层级	优化方向
IoU	像素级区域重叠	↑ 越高越好
F1	像素级区域一致性	↑ 越高越好
Pd	目标级检出率	↑ 越高越好
Fa	像素级背景误报率	↓ 越低越好

metrics.py

```python
#!/usr/bin/env python
# -*- coding:utf-8 -*-

"""
-------------------------------------------------
File Name: metrics.py
Description :Document description
Author: hzhou
date: 2026/11/3
-------------------------------------------------
Change Activity:
2026/11/3: Description of change
-------------------------------------------------
"""

import numpy as np
import cv2


class AS1000Metrics:
    def __init__(self, match_distance=3, connectivity=8):
        """
        Args:
            match_distance: 目标中心匹配距离阈值（像素）
            connectivity: 连通区域邻接方式（4 或 8）
        """
        self.match_distance = match_distance
        self.connectivity = connectivity
        self.reset()

    def reset(self):
        self.tp = 0
        self.fp = 0
```